MILESTONE - 1 PROJECT REPORT

AI-Based Patient Appointment System

Existing Healthcare Appointment Systems & AI Chatbot Study

PROJECT NAME	SUBMISSION MILESTONE
AI Patient Appointment System	**Milestone 1**
AUTHOR / STUDENT	ROLL NUMBER / ID
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GITHUB REPOSITORY	DATE OF SUBMISSION
LokendraDev06 / Repo Link	**August 2026**

1 Milestone 1 Study: Existing Healthcare & AI Chatbot Analysis

In recent years, digital health platforms have transformed patient care access. However, a comprehensive analysis of existing healthcare platforms reveals a structural divide: traditional booking platforms focus purely on calendar scheduling without diagnostic guidance, whereas advanced AI diagnostic chatbots operate in isolation without appointment integration.

1.1 Comparative Matrix of Existing Systems

Platform	Category	Core Capabilities	AI / LLM Level	Key Gaps & Limitations
Practo	BOOKING ENGINE	Doctor search, clinic directory, online consultation & slot booking.	LOW (KEYWORD SEARCH)	Patients must self-diagnose and select specialist; no preliminary AI triage.
Zocdoc	DOCTOR DISCOVERY	Location-based doctor discovery, insurance filter, real-time slots.	LOW (RULE SEARCH)	No symptom checker; requires patient to know their exact medical condition.
Ada Health	AI SYMPTOM CHECKER	Probabilistic disease assessment, risk factors, triage report.	HIGH (BAYESIAN NETWORK)	Does **not** offer appointment booking or direct doctor scheduling.
Babylon Health	TELEHEALTH AI	AI symptom assessment chatbot, virtual consultations, medical records.	HIGH (NLP & EXPERT AI)	High operational complexity, restricted clinic network integration.
Buoy Health	TRIAGE NAVIGATOR	Interactive triage questionnaire, level of care routing (ER vs Urgent).	MODERATE-HIGH (ML)	External referral links are fragmented; lacks unified booking database.

1.2 Major Gaps Identified

- **Triage & Booking Disconnect**
  Symptom evaluation platforms (Ada) and doctor booking portals (Practo) run independently, forcing users to switch apps manually.
- **Patient Self-Diagnosis Burden**
  Patients frequently select incorrect specialties (e.g. visiting Orthopedics for rheumatic joint pain), leading to wasted consultations.
- **Lack of Emergency Triage**
  Standard booking platforms lack instant detection of acute emergency symptoms like heart attacks or strokes, delaying urgent care.
- **Doctor Context Blindness**
  Doctors receive no prior structured summary of patient symptoms before consultation, spending valuable time on basic intake.

2 Problem Statement & System Objectives

PROBLEM STATEMENT

"Current healthcare scheduling portals require patients to self-diagnose and select medical specialists without clinical guidance, while AI symptom checkers operate in isolation without offering direct appointment booking. There is a critical need for an integrated system that combines AI-driven symptom assessment, automated doctor-specialty recommendation, emergency risk filtering, real-time slot scheduling, and structured clinical summaries for doctors."

Core System Objectives

- **Intelligent Symptom Triage:** Provide a 24/7 AI-powered conversational chatbot to collect patient symptoms, duration, and severity using natural language processing (NLP).
- **Automated Doctor-Specialty Mapping:** Map patient symptoms algorithmically to appropriate medical specializations (Cardiology, Dermatology, Neurology, etc.).
- **Emergency Risk Detection:** Instantly detect critical life-threatening symptoms and alert patients to seek emergency care (ER) immediately.
- **Seamless Slot Booking:** Enable real-time calendar slot reservation with verified doctors based on specialization, rating, and location.
- **Doctor Clinical Dashboard:** Generate automated pre-consultation AI clinical summaries for doctors to optimize consultation time.

3 Technology Stack

The system is designed with a modern, decoupled architecture separating the client interface, API gateway, AI recommendation logic, and data layer.

FRONTEND STACK

- **Framework:** React.js 18 + Vite
- **UI Design:** Modern CSS3, Glassmorphism, CSS Variables
- **Icons & Components:** Lucide-React Icons
- **HTTP Client:** Fetch API / Axios

BACKEND STACK

- **Runtime:** Python 3.14
- **Framework:** FastAPI (Asynchronous REST API)
- **Server:** Uvicorn ASGI Server
- **Validation:** Pydantic Schemas

AI ENGINE & ALGORITHMS

- **Emergency Classifier:** High-risk keyword matching
- **Symptom Matcher:** Set Overlap & TF-IDF similarity matcher
- **Knowledge Base:** Medical dataset mapping symptoms to disease & specialty

DATABASE & ARCHITECTURE

- **Dataset Storage:** CSV / JSON Medical Mapping
- **Data Models:** In-Memory DB (Milestone 1) -> SQLite / PostgreSQL (Milestone 3)
- **Security:** CORS middleware, input sanitization

4 System Architecture, ER Diagram & Process Flowchart

4.1 System Architecture Diagram

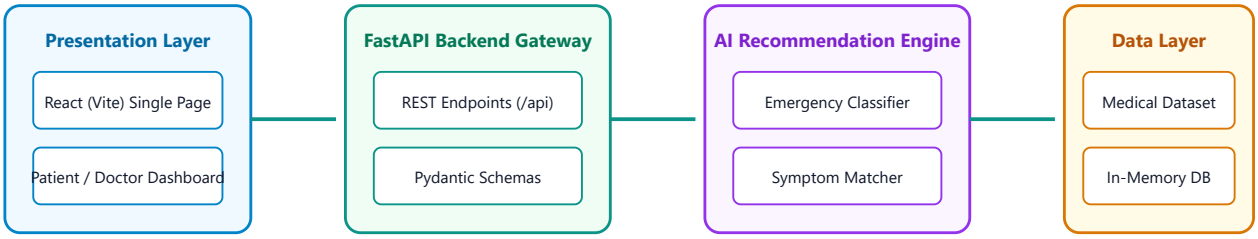


FIGURE 1: HIGH-LEVEL TIERED SYSTEM ARCHITECTURE

4.2 Entity-Relationship (ER) Diagram

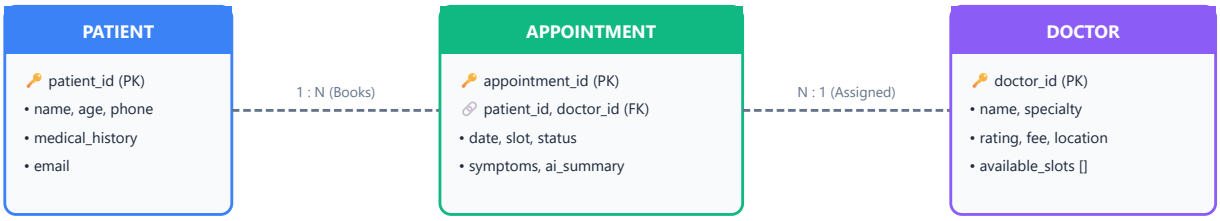


FIGURE 2: ENTITY-RELATIONSHIP DIAGRAM FOR CORE DATA MODELS

4.3 Process Flowchart

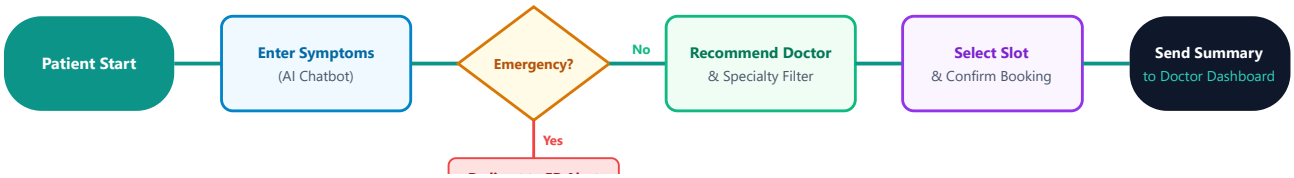


FIGURE 3: END-TO-END SYSTEM WORKFLOW FLOWCHART

5 Milestone 1 Deliverables & Current Progress

During Milestone 1, the core frontend interface modules, RESTful API endpoints, and AI assessment logic have been successfully developed and integrated into a functional working system.

5.1 Implemented Frontend Modules

Module Name	File Location	Key Functionality	Status
Navbar	`src/components/Navbar.jsx`	Header navigation, branding, quick role switcher (Patient / Doctor View).	COMPLETED
Intro Onboarding	`src/components/IntroOnboarding.jsx`	Hero section, system highlights, emergency disclaimer, chatbot triggers.	COMPLETED
AI Chatbot Drawer	`src/components/ChatbotDrawer.jsx`	Interactive symptom assistant with real-time API assessment responses.	COMPLETED
Doctor Cards	`src/components/DoctorCard.jsx`	Displays filtered doctor profiles, ratings, specialization & slot selection.	COMPLETED
Appointment Modal	`src/components/AppointmentModal.jsx`	Date picker, slot selection, and direct booking API integration.	COMPLETED
Doctor Dashboard	`src/components/DoctorDashboard.jsx`	Doctor schedule viewer, patient list & AI clinical summary view.	COMPLETED

5.2 Implemented Backend Endpoints

Method	Endpoint	Description	Response payload
GET	`/`	Health check & system operational status	`{"status": "Online"}`
POST	`/api/chat/assess`	Symptom evaluation & emergency check	Matched disease, recommended specialist, summary
GET	`/api/doctors`	Fetch doctor list (optional specialty filter)	Array of doctor objects & available slots
POST	`/api/appointments/book`	Reserve slot & record AI clinical summary	Booking confirmation & appointment ID
GET	`/api/appointments/doctor/{id}`	Retrieve doctor's patient queue & summaries	List of confirmed patient appointments

6 GitHub Repository & Live Demo Details

📌 **Official GitHub Repository**

URL: https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System

The repository contains complete frontend React source code, FastAPI backend application, medical datasets, documentation, and Milestone 1 report files.

Local Run Instructions

1. **Frontend:** Navigate to `/frontend`, run `npm install` then `npm run dev` (Server runs at `http://localhost:5173`).
2. **Backend API:** Navigate to `/backend`, run `pip install -r requirements.txt` then `python app/main.py` (API runs at `http://localhost:8000`).
3. **Interactive Swagger Docs:** Accessible at `http://localhost:8000/docs`.